

KOLOBRZEG, Poland

WMO: 121000

Lat: 54.1825N

Lon: 15.5797E

Elev: 18

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	17.7	5.8	7.4	14.5	10.9	9.6	18.7	34.4	39.5	31.3	38.5	4.9	140	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	79.4	66.5	75.5	65.9	72.7	64.8	69.2	75.9	67.5	73.5	65.9	71.1	6.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.7	98.5	72.4	65.2	93.1	70.6	63.8	88.6	69.1	33.3	76.2	31.9	73.9	30.7	71.3	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.2	22.7	18.5	DB	10.1	88.7	6.7	4.6	5.3	92.0	1.4	94.7	-2.4	97.3	-7.3	100.7	
			WB	9.1	72.1	6.5	1.8	4.5	73.4	0.7	74.4	-2.9	75.4	-7.6	76.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	32.7	33.6	38.0	46.0	53.5	60.4	64.0	64.2	58.6	50.1	42.5	36.9
	DBStd	12.74	8.40	7.13	6.71	5.97	6.38	5.27	4.00	4.51	4.46	5.54	5.33	7.09
	HDD50	2254	536	458	374	151	32	1	0	0	1	67	229	405
	HDD65	6146	1001	878	837	570	363	159	67	68	196	461	676	870
	CDD50	1694	0	0	3	31	140	311	435	441	259	71	3	0
	CDD65	111	0	0	0	0	6	20	37	43	5	0	0	0
	CDH74	531	0	0	0	4	70	136	108	191	22	0	0	0
	CDH80	127	0	0	0	0	13	41	15	57	1	0	0	0
Wind	WSAvg	7.7	8.5	8.3	8.0	7.9	7.4	7.4	7.2	6.7	7.0	7.1	7.8	9.0
Precipitation	PrecAvg	27.8	1.9	1.6	1.8	1.4	2.1	2.8	3.0	3.5	2.8	2.4	2.2	2.2
	PrecMax	36.0	4.9	2.7	3.9	3.3	4.0	7.6	6.5	7.9	6.0	5.9	4.3	4.2
	PrecMin	23.0	0.1	0.2	0.4	0.3	0.5	0.3	0.2	0.4	0.7	0.7	0.2	0.7
	PrecStd	3.4	1.1	0.6	0.9	0.7	0.8	1.4	1.6	1.6	1.3	1.3	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	52.0	60.7	71.3	82.1	85.2	81.8	86.8	77.4	67.8	56.3	53.1
		MCWB	48.9	47.1	49.9	56.7	64.7	67.8	68.6	68.8	63.4	58.4	53.3	50.0
	2%	DB	48.0	48.0	53.5	63.8	73.9	77.2	76.5	78.1	72.7	63.8	53.7	50.2
		MCWB	45.5	44.2	48.5	52.0	61.5	65.8	67.8	67.6	62.1	57.9	50.7	47.6
	5%	DB	45.2	45.2	49.9	58.7	67.8	72.0	74.0	74.3	68.9	61.0	52.0	47.5
		MCWB	43.0	42.0	45.5	50.3	58.8	63.6	66.9	66.1	61.4	56.4	49.5	44.9
	10%	DB	42.9	42.5	46.9	54.8	63.1	68.1	71.5	71.5	66.0	58.3	50.3	45.1
		MCWB	40.8	39.9	43.4	48.6	56.7	62.0	65.5	64.8	60.1	54.8	48.1	42.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	48.7	52.5	56.8	65.7	69.3	71.8	71.6	66.9	60.6	54.3	50.8
		MCDB	52.2	50.0	57.5	66.5	77.1	79.1	78.5	80.8	72.4	64.7	55.8	52.5
	2%	WB	45.7	44.6	48.7	53.4	62.7	66.7	69.5	69.0	63.9	58.7	51.4	48.0
		MCDB	47.8	47.4	52.8	61.2	72.8	76.4	74.7	76.2	68.8	62.7	53.1	50.1
	5%	WB	43.3	42.2	45.9	51.2	59.8	64.4	67.7	67.3	62.4	56.8	49.8	45.2
		MCDB	45.1	44.9	49.5	57.7	66.6	70.3	72.5	73.1	67.1	60.2	51.6	47.2
	10%	WB	41.1	40.0	43.6	49.2	57.2	62.6	66.1	65.7	61.0	55.1	48.2	43.2
		MCDB	42.7	42.3	46.4	54.6	62.6	67.6	70.2	70.6	65.4	57.9	50.2	44.8
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.4	9.6	11.2	12.1	11.6	11.0	12.5	12.0	10.2	7.0	6.3
		MCDBR	7.1	10.6	14.8	19.2	22.4	19.6	16.7	17.8	17.5	14.1	8.4	8.0
		MCWBR	6.6	8.0	9.6	11.4	12.3	10.7	9.7	9.8	9.9	9.4	6.9	7.2
	5% WB	MCDBR	7.1	9.8	13.1	16.9	19.6	16.7	14.5	15.6	14.1	12.6	8.0	7.9
		MCWBR	6.7	8.0	9.2	10.9	11.7	10.2	9.4	9.6	9.5	9.4	7.0	7.4
Clear-Sky Solar Irradiance	taub	0.317	0.342	0.369	0.406	0.405	0.399	0.423	0.421	0.383	0.364	0.343	0.314	
	taud	2.304	2.308	2.274	2.239	2.307	2.352	2.299	2.313	2.395	2.390	2.343	2.307	
	Ebn at Noon	199	234	254	259	267	270	260	252	248	224	187	174	
	Edh at Noon	19	26	33	39	38	37	38	36	29	23	18	16	
All-Sky Solar Radiation	RadAvg	154	359	763	1337	1718	1823	1669	1381	995	503	200	108	
	RadStd	19	56	72	159	194	131	198	115	103	74	27	11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-238	N/A	N/A

Nomenclature: See separate page